No. $\therefore P(X=k) = \frac{e^{-\lambda}\lambda^k}{k!}$

Date.

Homework #5

习题四 T17, 18, 21, 27, 29

T17

	鲫鱼	鳊鱼	
0~1kg	$\lambda_{11}=1$	$\lambda_{12}=\frac{1}{2}$	$\lambda_3=\frac{3}{2}$
1~2kg	$\lambda_{21}=1$	$\lambda_{22}=\frac{1}{2}$	$\lambda_4=\frac{3}{2}$
	$\lambda_1=2$	$\lambda_2=1$	

(1) $\lambda=\lambda_1+\lambda_2=3$ $N(1)\sim\pi(3)$

$P(N(1)=2)=\frac{9}{2}e^{-3}$

(2) $\lambda_3=\frac{3}{2}$, $\lambda_4=\frac{3}{2}$, $N_3(1)\sim\pi(\frac{3}{2})$, $N_4(1)\sim\pi(\frac{3}{2})$

$P(N_3(1)=2, N_4(1)=2)=\frac{81}{64}e^{-3}$

(3) $\lambda_{21}=1$, $P(N_{21}(1)=1, N_{21}(2)=2, \tilde{N}_{21}(1)=1, \tilde{N}_{21}(2)=2)$

$= e^{-1}e^{-1}e^{-2}e^{-2}=e^{-6}$

(4) $P(N_{21}(2)=2, \tilde{N}_{21}(2)=0 \mid N(2)=2)$

$=\frac{P(N_{21}(2)=2, \tilde{N}_{21}(2)=0)}{P(N(2)=2)} = \frac{\frac{e^{-2}2^2}{2}\frac{e^{4}4^0}{1}}{\frac{e^{-6}6^2}{2}} = \frac{1}{9}$

$Cov(B(s), B(t)) = \min(s,t)$

$N(t) - N(s) \sim \lambda(\int_s^t \lambda(u) du)$ $B(t) - B(s) \sim N(0, t-s)$

$F(x) = P(X \le x) = \phi(\frac{x-\mu}{\sigma})$

T18 (1) $N(2) \sim \lambda(\int_0^2 t\,dt) = \lambda(2)$

$P(N(2)=3) = \frac{4}{3}e^{-2}$

(2) $P(N(1)=2, N(2)=4)$

$= P(N(1)=2, N(2)-N(1)=2)$

$N(1) \sim \lambda(\int_0^1 t\,dt) = \lambda(\frac{1}{2})$ $N(2)-N(1) \sim \lambda(\int_1^2 t\,dt) = \lambda(\frac{3}{2})$

$P = \frac{1}{8}e^{-\frac{1}{2}} \cdot \frac{9}{8}e^{-\frac{3}{2}} = \frac{9}{64}e^{-2}$

(3) $P(N(1)=2 \mid N(2)=4)$

$= \frac{P(N(1)=2, N(2)-N(1)=2)}{P(N(2)=4)} = \frac{\frac{9}{64}e^{-2}}{\frac{2}{3}e^{-2}} = \frac{27}{128}$

$N(2) \sim \lambda(\int_0^2 t\,dt) = \lambda(2)$

T21 (1) $P(B(3.6) \le 1 \mid B(1.6)=0.8, B(2.39)=-0.1)$

$= P(B(3.6)-B(2.39) \le 1.1)$

$= \phi(1)$

$B(3.6)-B(2.39) \sim N(0, 1.21)$

(2) $Cov(B(8)-B(4), B(6))$

$= Cov(B(8), B(6)) - Cov(B(4), B(6)) = 6-4 = 2$

(3) $D(2B(1)+B(2)) = 4D(B(1)) + 4Cov(B(1), B(2)) + D(B(2))$

$= 10$

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No. $B(t) = t\tilde{B}(\frac{1}{t})$　　为什么正算是错的

Date.

T27 (1) $P(B(\frac{1}{10}) \geq \frac{3}{2} \mid B(\frac{1}{6}) = 2, B(\frac{1}{4}) = \frac{12}{5})$

$= P(\frac{1}{10}\tilde{B}(10) \geq \frac{3}{2} \mid \frac{1}{6}\tilde{B}(6) = 2, \frac{1}{4}\tilde{B}(4) = \frac{12}{5})$

$= P(\tilde{B}(10) \geq 15 \mid \tilde{B}(6) = 12, \tilde{B}(4) = \frac{48}{5})$

$= P(\tilde{B}(10) - \tilde{B}(1) \geq 3)$

$= 1 - \phi(\frac{3}{2})$

$\tilde{B}(10) - \tilde{B}(6) \sim N(0, 4)$

(2) 即求 $\tilde{B}(6) = 12$, $\tilde{B}(4) = \frac{48}{5}$ 时 $\frac{1}{10}\tilde{B}(10)$ 的条件分布

$\tilde{B}(10) \sim 12 + N(0, 4)$　　$\frac{1}{10}\tilde{B}(10) \sim N(1.2, 0.04)$

T29 (1) $B(t) \sim N(0, t)$

$P(|B(t)| \leq x) = P(-x \leq B(t) \leq x) = 2\phi(\frac{x}{\sqrt{t}}) - 1$

(2) $P(\max_{0 \leq s \leq t} B(s) - B(t) \leq x)$

$= P(\max_{0 \leq s \leq t}(B(s) - B(t)) \leq x)$

令 $s = t - \mu$, 则 $P = P(\max_{0 \leq \mu \leq t}(B(\mu) - B(0)) \leq x)$

$= 1 - 2P(B(t) > x)$

$= 2\phi(\frac{x}{\sqrt{t}}) - 1$

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